



TECHNOLOGICAL
UNIVERSITY(MANDALAY)
DEPARTMENT OF CIVIL ENGINEERING

TUM GRAND RESIDENT HOUSING PROJECT WITH VISIBLE FLOOR LEVEL

Introduction

- TUM Grand Residences project is a residential development study.
- The project includes a shopping mall, two-storey and three-storey buildings.
- It focuses on building planning, layout design, and supporting facilities.
- Hydraulic jack principle and installation are also explained.
- The project aims to understand practical engineering applications

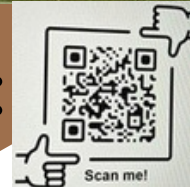


Objectives of the Study

- 1.To study the layout and planning of a modern residence.
- 2.To develop skills in architectural model making.
- 3.To gain knowledge and practical experience in using AutoCAD, SketchUp & Revit software.
- 4.To improve creativity, accuracy, and problem-solving skills.
- 5.To gain experience in project planning and teamwork.
- 6.To improve presentation skills and explain the residence design effectively.



SCAN ME:



THE MAIN MODEL IS
BUILT AT A 3:8 SCALE
WITH A HYDRAULIC
JACK SYSTEM FOR
LIFTING AND
ADJUSTMENT.

ABSTRACT

This project focuses on constructing a scaled model of a residential villa development with a shopping mall, multi-storey buildings, a gate system, and a hydraulic lift.

It demonstrates architectural planning, structural organization, load distribution, and mechanical integration using materials such as foam board and acrylic sheets. The completed model, including detailing, painting, and landscaping, provides practical knowledge of architectural design, structural principles, hydraulic systems, and model-making techniques.

A 1:16 SCALE IS USED FOR THE SMALLER HOUSE MODELS INSIDE THE HOUSING COMPOUND.➡



AIM:

To study and demonstrate the planning, design, modelling, and construction concepts of a residential development, including the practical application of a hydraulic jack system using Pascal's Law.

